Exercise 1

Table from the task 2

	Total number of cars	Cars per minute
Traffic_Laramie_1.mp4	6	3
Traffic_Laramie_2.mp4	4	3

Frame differencing and background substruction techniques

Frame differencing and background substruction techniques are techniques that are based on analyzing the difference between two different images, also referred to as frames.
The functionality is based on taking same pixel from both images and compare if there is a difference between them. Neglecting all the pixels that are the same or very similar we can then have set of pixels that are different, and that allows us to track movement.

Frame differencing refers to that technique of neglecting pixels that are the same in both images, and showing which pixels aren't.
Background substruction technique is using frame differencing but, in this idea, always one of the images that are being compared for frame differencing is an image where every pixel is part of the background, and then all the other frames are always compared to this background image.

Analysis of application

Task 1:

- Use CV to analyze file data
- Loop over every frame of the video, to make algorithm more efficient first, change frame to gray, then blur it, save that frame as background frame if its first frame,
- Use cv function to find frame difference between current frame and a background frame, this way we can subtract the background from the current frame and set contours
- Loops over contours and filters the contours that are too small to be a car, and draws green rectangle around the contours that are big enough to be a car
- Play the video frame by frame, close the feed when there are no more frames or if user presses q to exit sooner

Task 2:

- Use CV to analyze file data
- Set points of interest since the street is two way I can just count cars if there is contour in certain spot on the screen since at this spot cars will always go in direction of downtown
- Loop over every frame of the video, again to maintain efficiency change it to just gray scale values so instead of having so many values per pixel there is one just scale of gray, and gaussian blur takes pixels around and makes one pixel that is average of that making less pixel to compare, save background frame if first frame
- find frame difference
- Loops over contours and filters the contours that are too small to be a car and if the contour is in point interest set car detected to true, to make sure single car is not counted twice, or frame glitch I added logic that there must be no car detected in the contour in that frame for 30 frames that gets insures accurate counting,
- Play video feed frame by frame and return amount of cars

- Another functions uses cv library to analyze duration of video

- Used math ceiling to divide number of cars per duration of video

- Use tabulate to present the results as a table